Diferença de Gaussianas

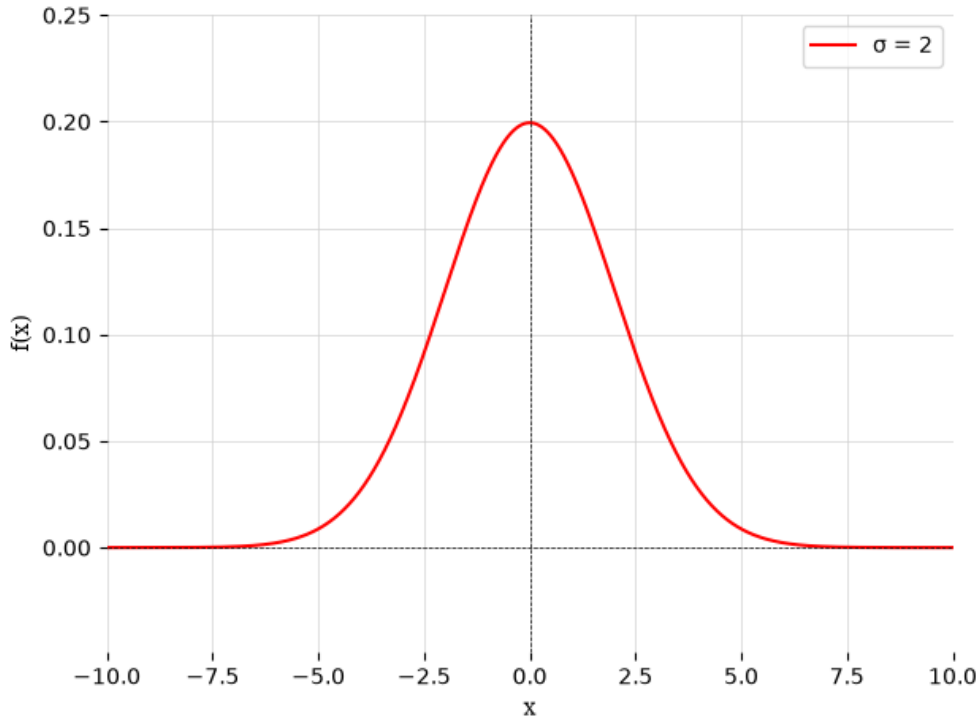
Tiago Maia de Lacerda Brasil

PPGC - IC/UFF

2026

Gaussian blur

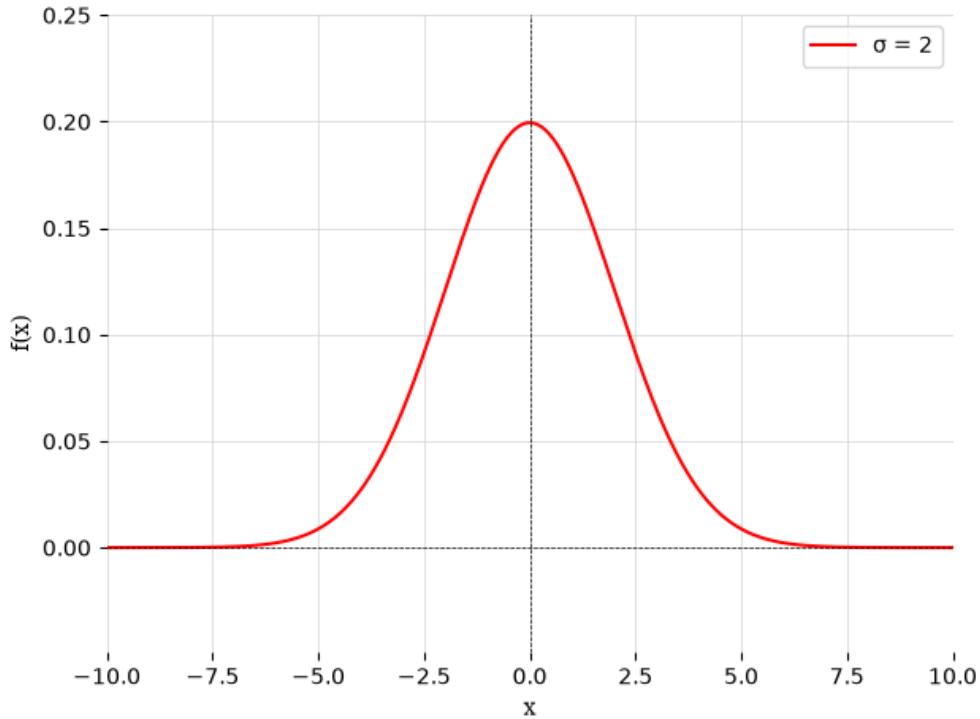
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Gaussian blur



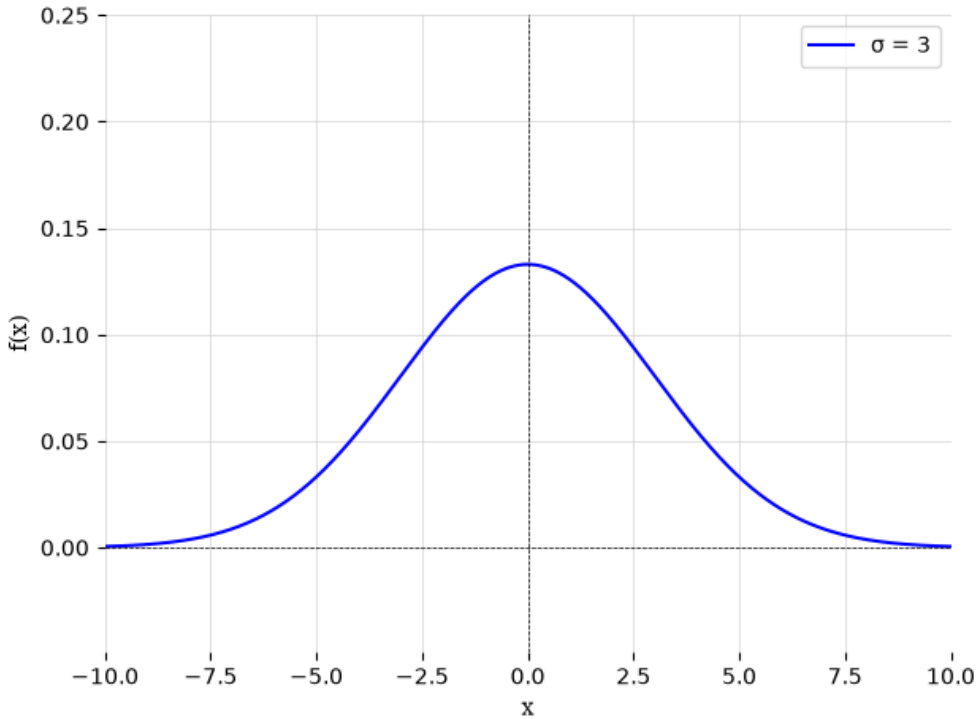
$\sigma = 2$



Gaussian blur



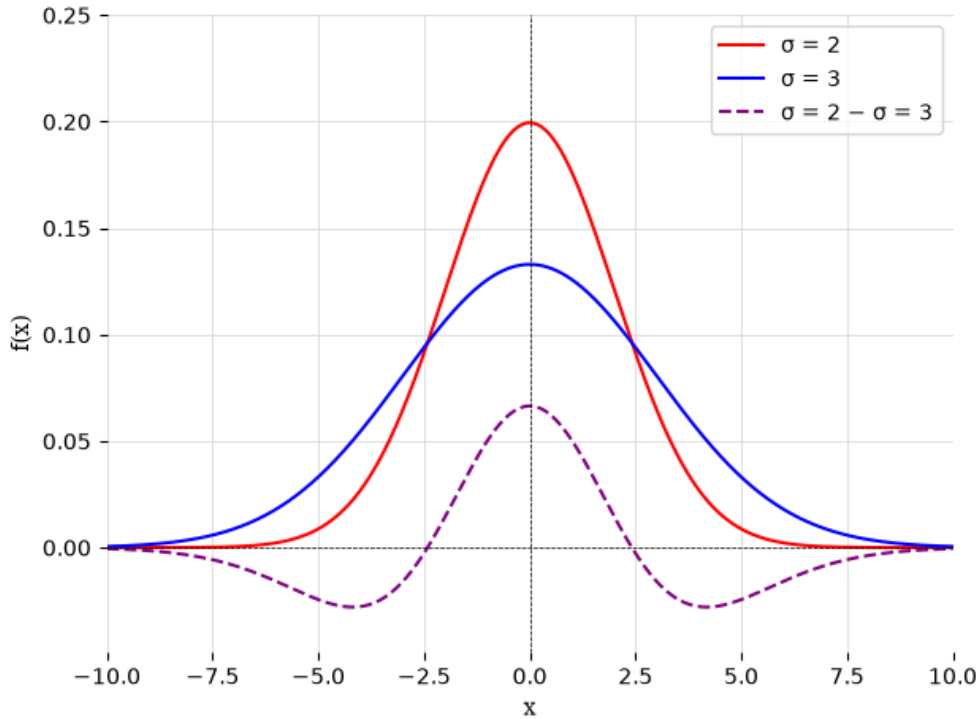
$\sigma = 3$



Gaussian blur



$(\sigma = 2) - (\sigma = 3)$



Sobel operator



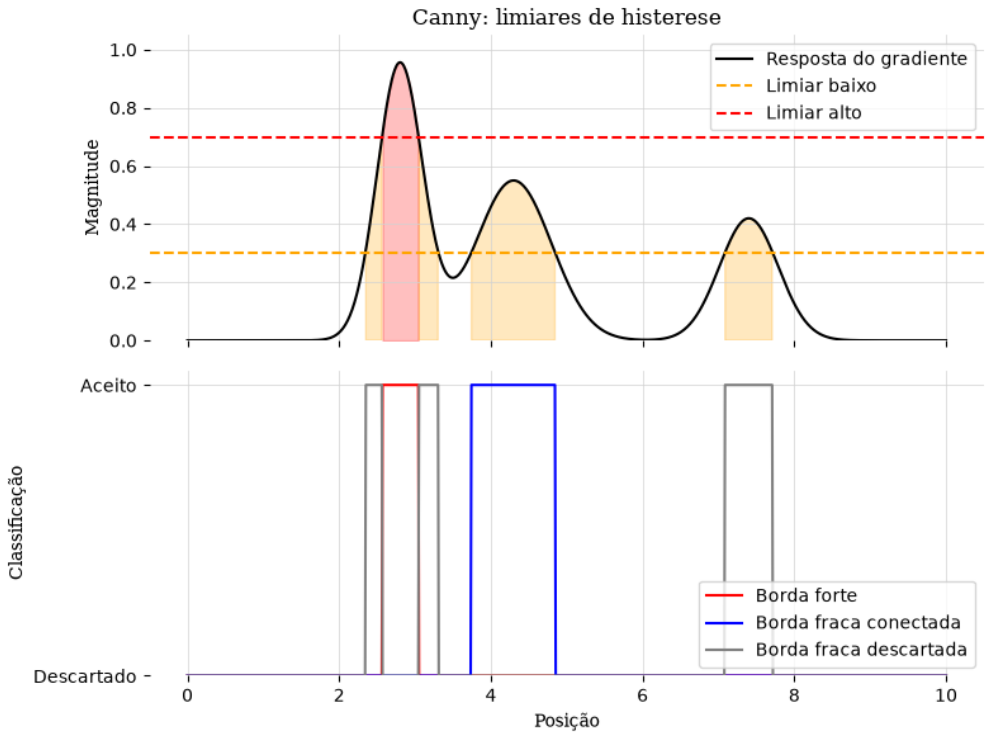
$$I = \sqrt{G_x^2 + G_y^2}$$

$$G_x = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -2 & 0 & 2 \\ \hline -1 & 0 & 1 \\ \hline \end{array} * I$$

$$G_y = \begin{array}{|c|c|c|} \hline -1 & -2 & -1 \\ \hline 0 & 0 & 0 \\ \hline 1 & 2 & 1 \\ \hline \end{array} * I$$

Canny edge detector





Por que pré-processar?

XDoG com os mesmos parâmetros



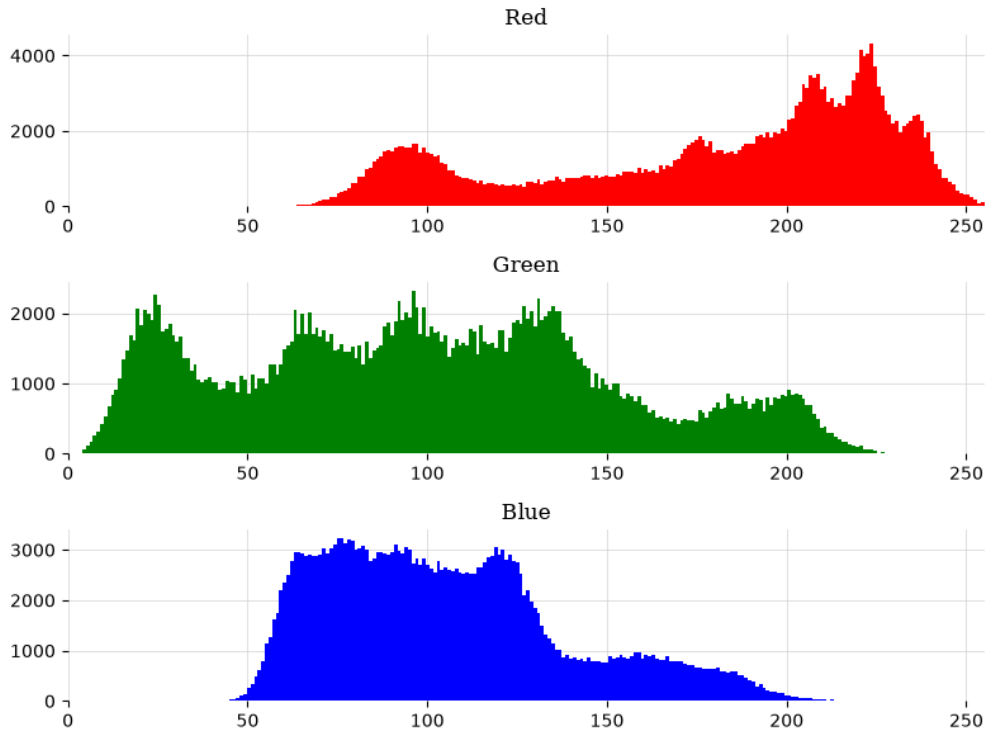
Y



$0{,}5\,Y$

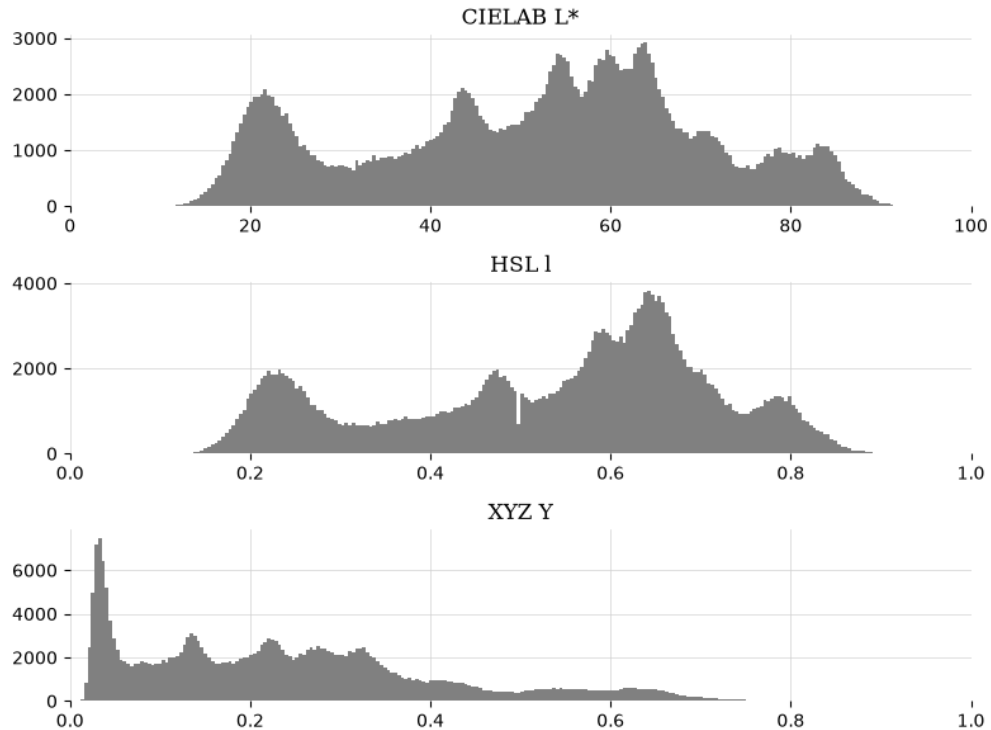
Espaços de cores





Espaços de cores





Espaços de cores

XDoG com os mesmos parâmetros



CIELAB L*

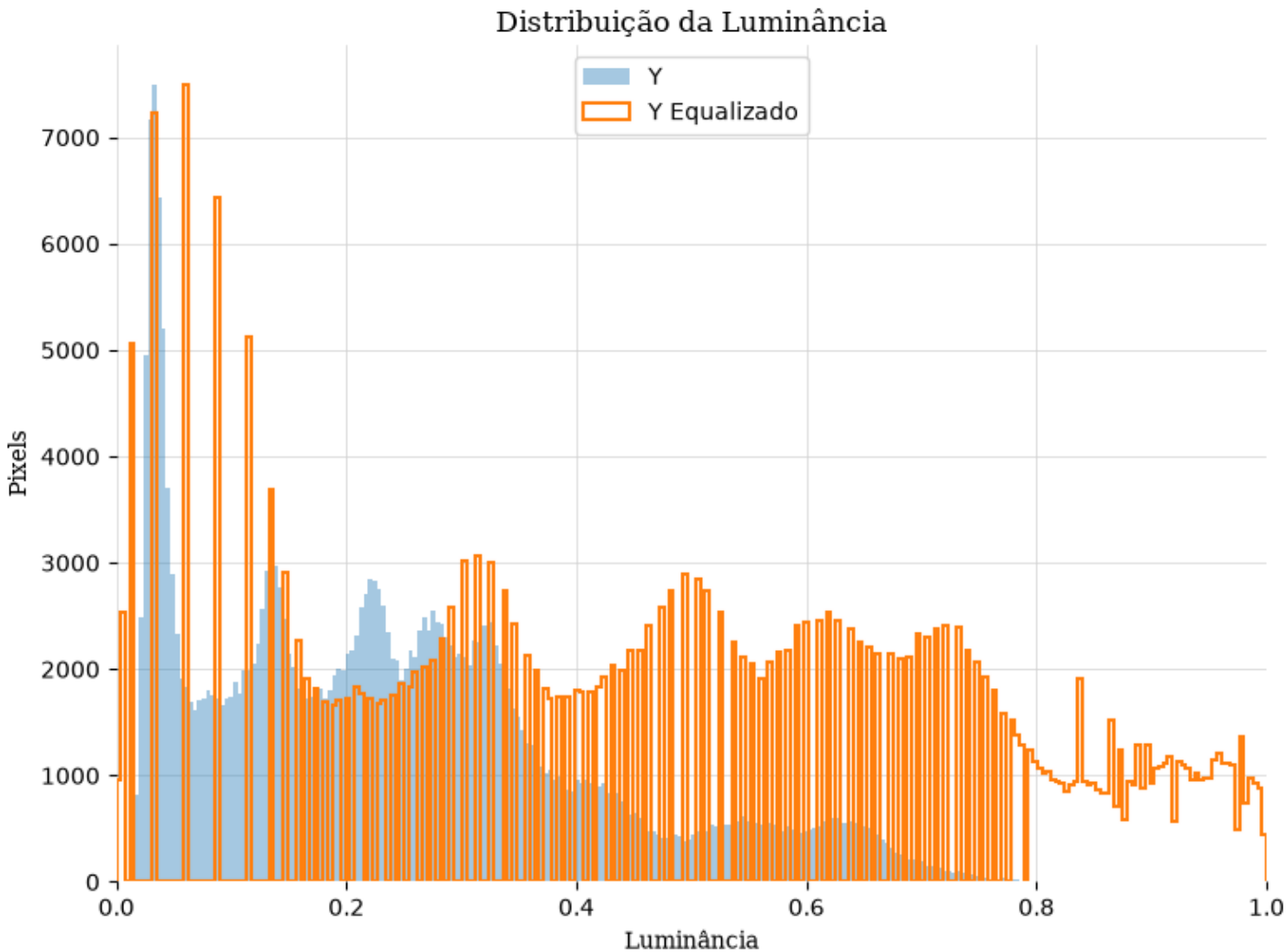


HSL l



XYZ Y

Balanceamento de contraste



Balanceamento de contraste



Y



Y equalizado

Balanceamento de contraste

XDoG com os mesmos parâmetros

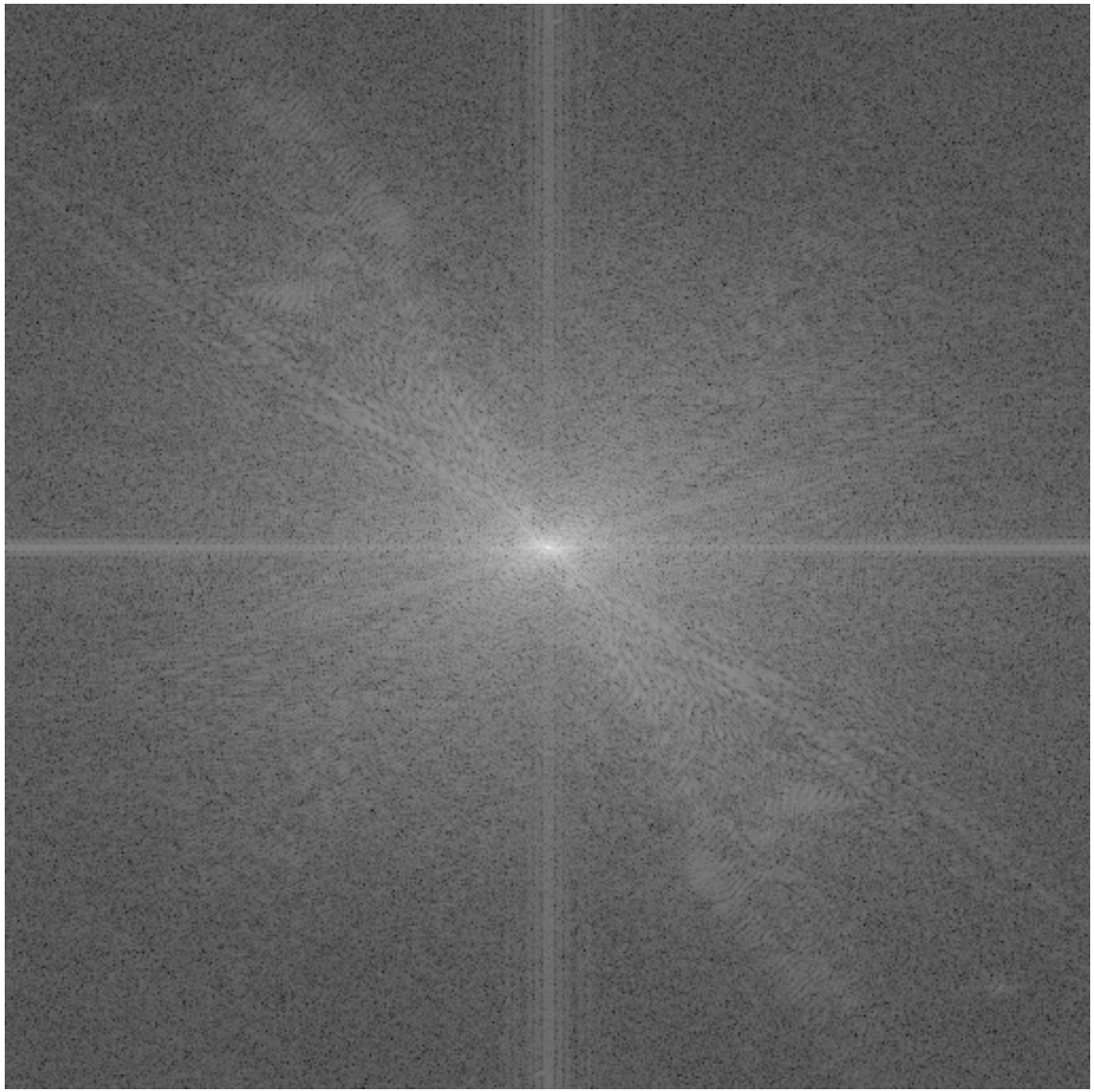


Y



Y equalizado

Análise espectral



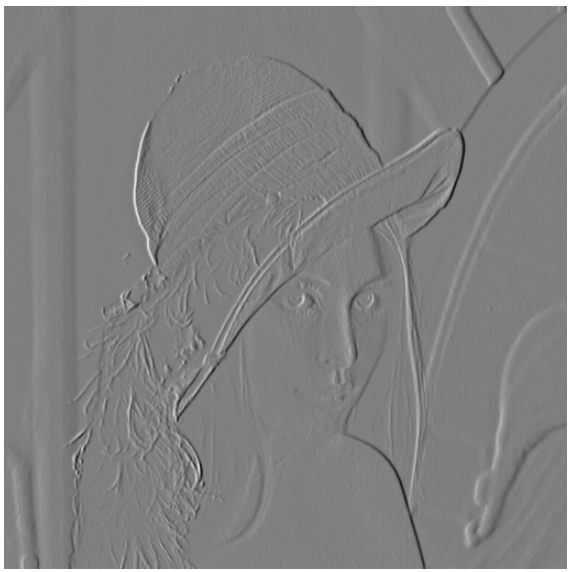
Diferença das Gaussianas Baseada em Fluxo

Imagem de entrada



Diferença das Gaussianas Baseada em Fluxo

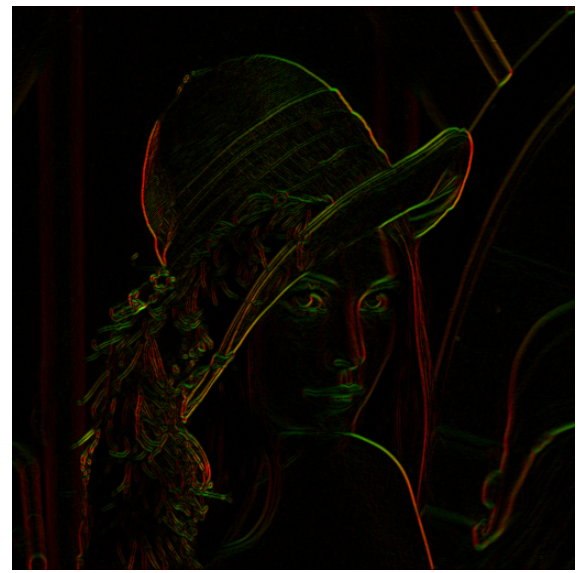
Gradientes de Sobel



G_x



G_y



$(|G_x|, |G_y|, 0)$

Diferença das Gaussianas Baseada em Fluxo

Tensor de estrutura

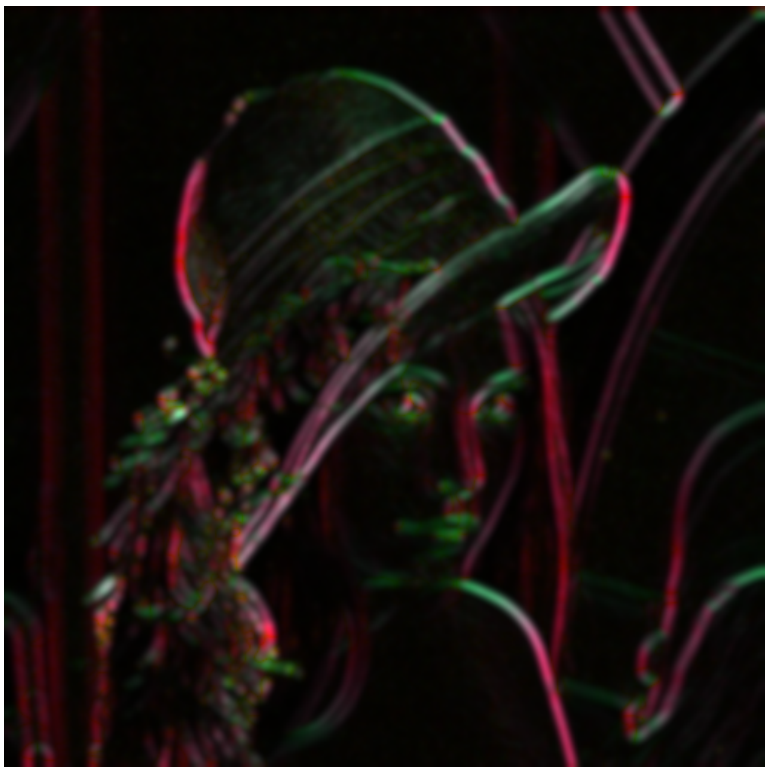
$$S = \begin{bmatrix} G_x * G_x & G_x * G_y \\ G_x * G_y & G_y * G_y \end{bmatrix} = \begin{bmatrix} E & F \\ F & G \end{bmatrix}$$

Diferença das Gaussianas Baseada em Fluxo

Suavização do tensor de estrutura



S



$S_{\sigma_c} = G_{\sigma_c} \otimes S, \quad \sigma_c = 2$

Cores: $(\sqrt{E}, \sqrt{G}, \sqrt{|F|})$

Diferença das Gaussianas Baseada em Fluxo

Vetor tangente

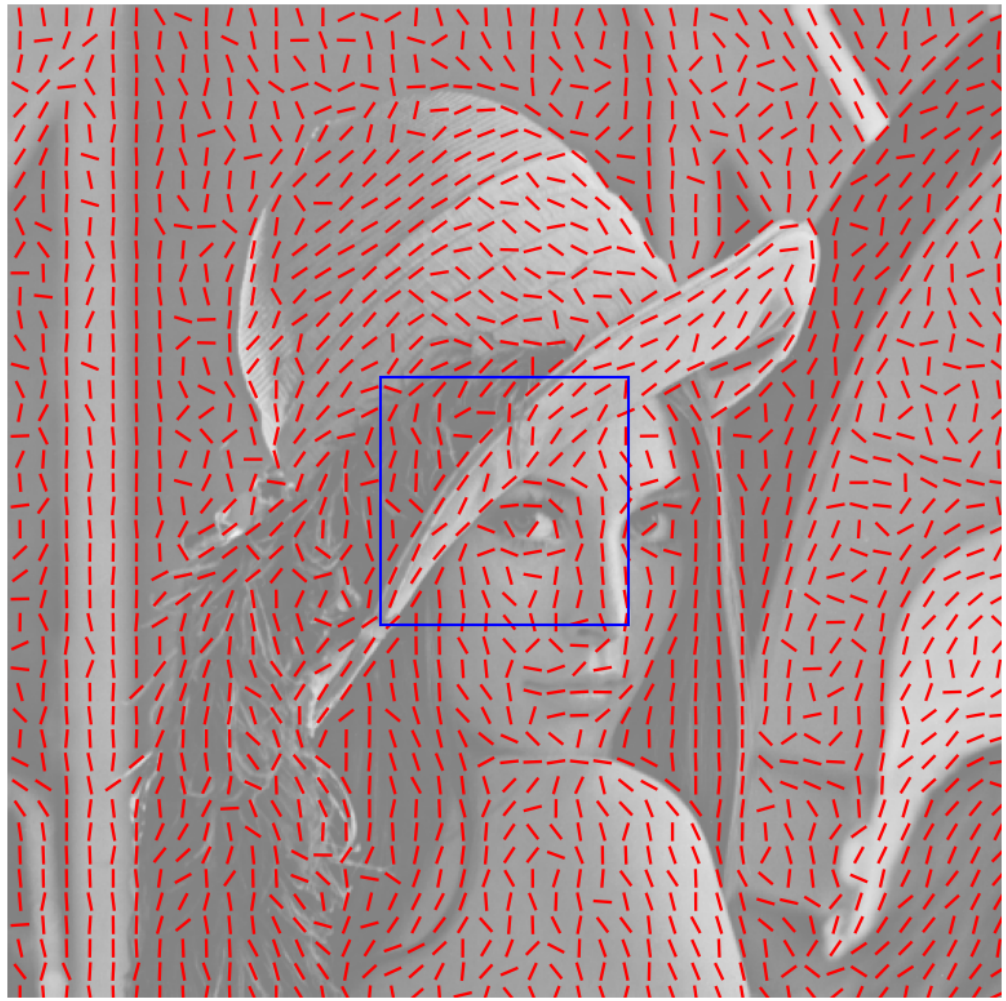
$$S_{\sigma_c} = G_{\sigma_c} \otimes S = \begin{bmatrix} \bar{E} & \bar{F} \\ \bar{F} & \bar{G} \end{bmatrix}$$

$$\lambda_1 = \frac{\bar{E} + \bar{G} + \sqrt{(\bar{E} - \bar{G})^2 + 4\bar{F}^2}}{2}$$

$$t = \begin{pmatrix} \lambda_1 - \bar{E} \\ -\bar{F} \end{pmatrix}, \qquad \hat{t} = \frac{t}{\|t\|}$$

Diferença das Gaussianas Baseada em Fluxo

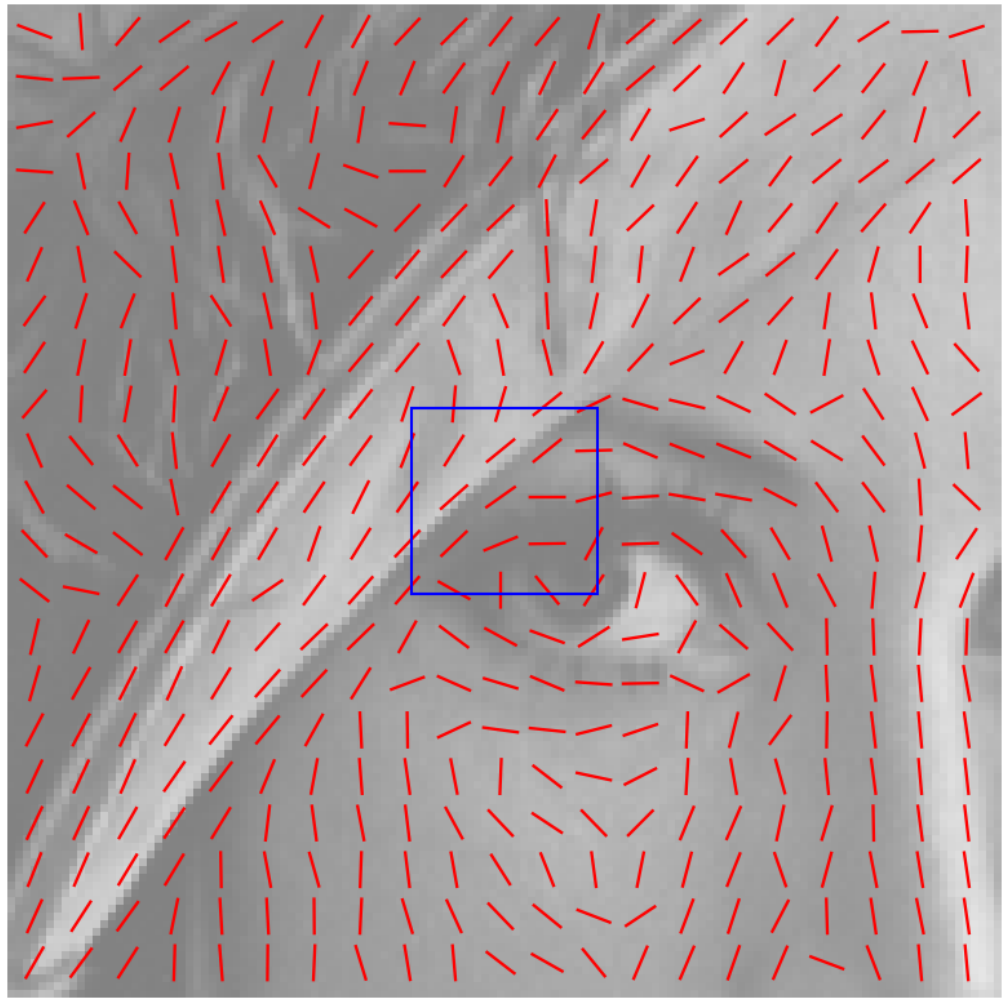
Campo de fluxo



Direção de menor variação $\hat{t}$, um segmento a cada 12 pixels

Diferença das Gaussianas Baseada em Fluxo

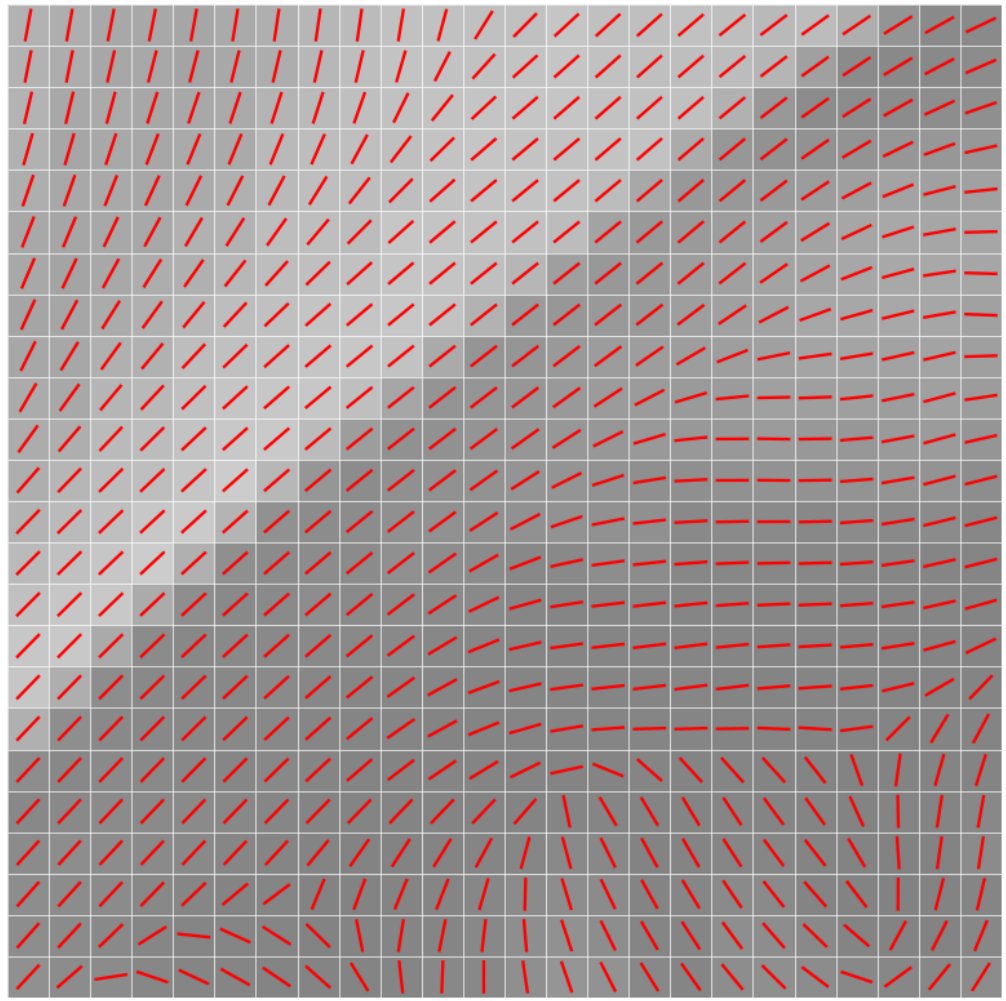
Campo de fluxo



Recorte central de 128 pixels, um segmento a cada 6 pixels

Diferença das Gaussianas Baseada em Fluxo

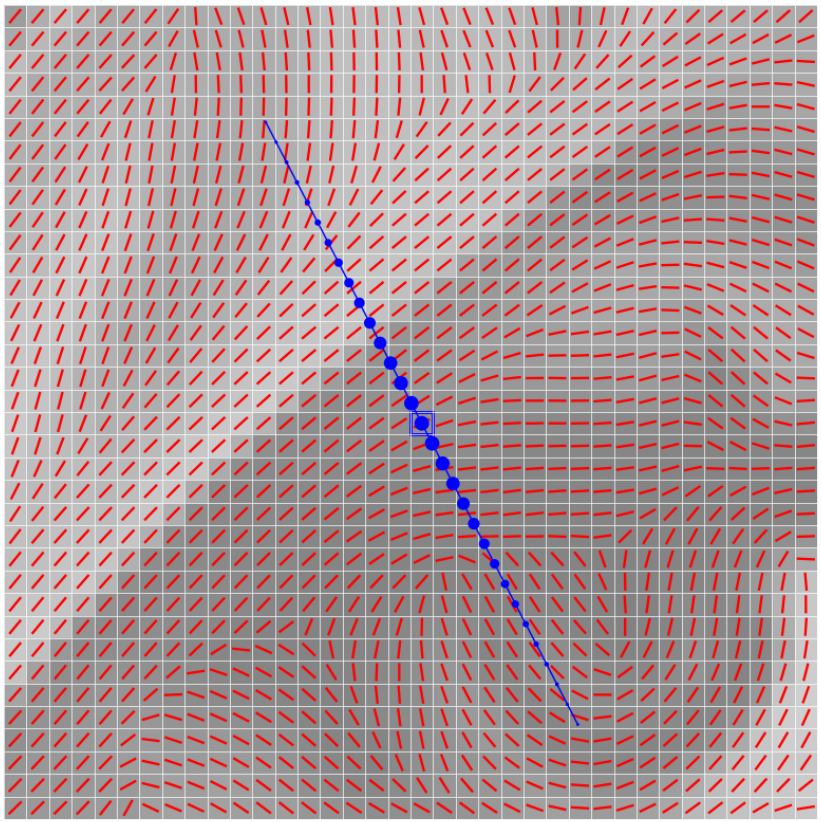
Campo de fluxo



Recorte central de 24 pixels, um segmento por pixel

Diferença das Gaussianas Baseada em Fluxo

Blur transversal



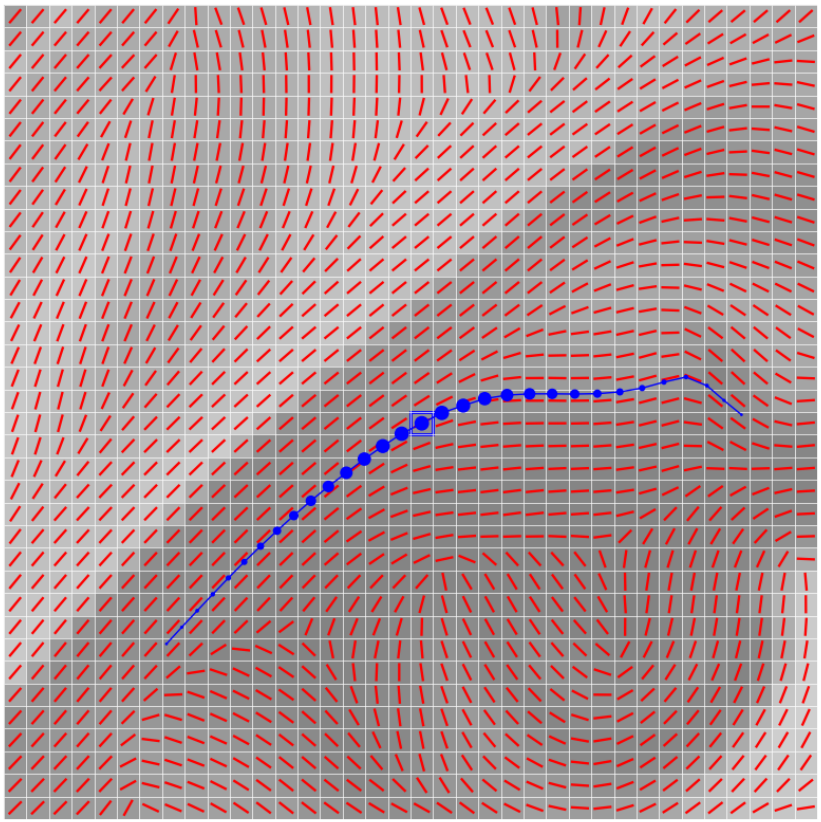
Amostras em linha reta, cruzando $\hat{t}$; tamanho é o peso



Resultado, $\sigma = 5$

Diferença das Gaussianas Baseada em Fluxo

Blur ao longo do fluxo (LIC)



Amostras ao longo da curva que segue $\hat{t}$; tamanho é o peso



Resultado, $\sigma = 5$

Diferença das Gaussianas Baseada em Fluxo

DoG transversal



Blur transversal



$$D = (1+p)\,G_\sigma - p\,G_{k\sigma}$$

$$\sigma = 1,\ k = 1{,}6,\ p = 20$$

Diferença das Gaussianas Baseada em Fluxo

Suavização ao longo do fluxo



DoG transversal D



$L(\mathbf{x}) = \sum_s G_{\sigma_m}(s)\, D(c_{\mathbf{x}}(s))$

$\sigma_m = 3$

Diferença das Gaussianas Baseada em Fluxo

Limiarização



Resposta suavizada L



$T = \min\left(1,\ 1 + \tanh(\varphi\,(L - \varepsilon))\right)$

$\varepsilon = 0{,}1,\ \varphi = 50$

Diferença das Gaussianas Baseada em Fluxo

Exemplos





Diferença das Gaussianas Baseada em Fluxo

Exemplos - *Pop Art*





Diferença das Gaussianas Baseada em Fluxo

Exemplos - *Posterization*





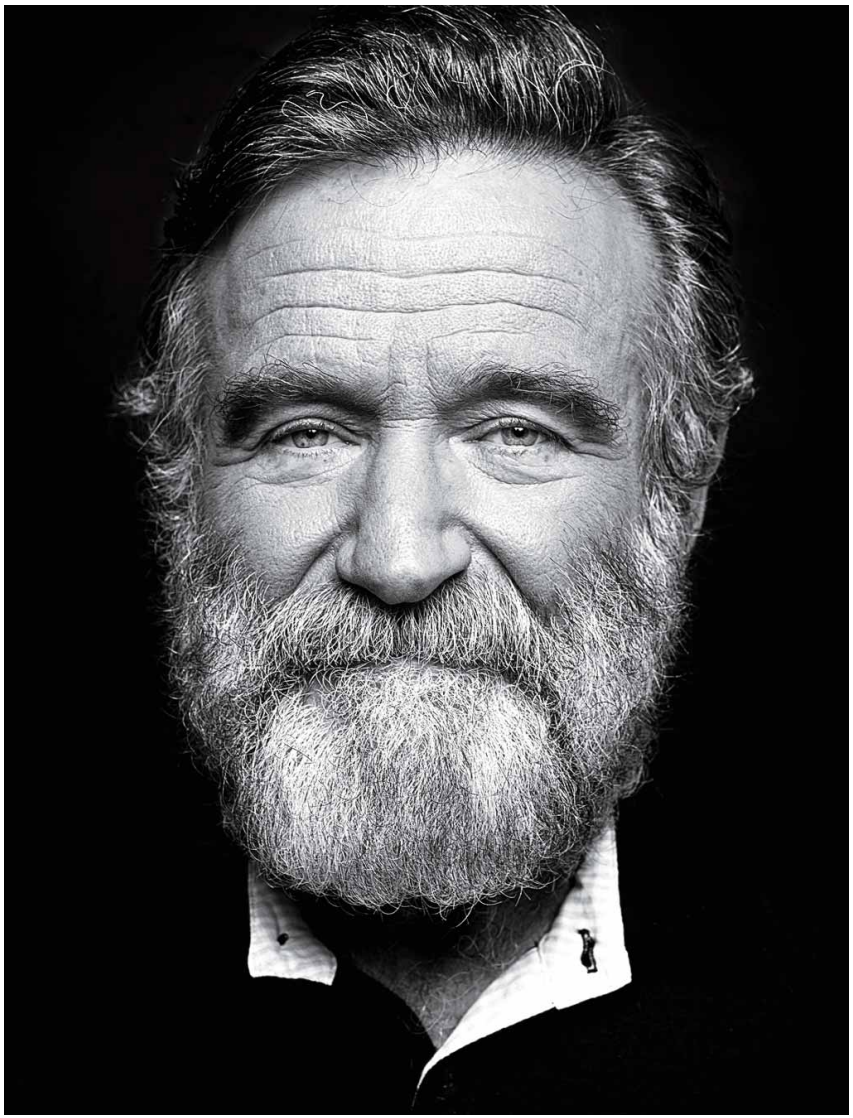
Diferença das Gaussianas Baseada em Fluxo

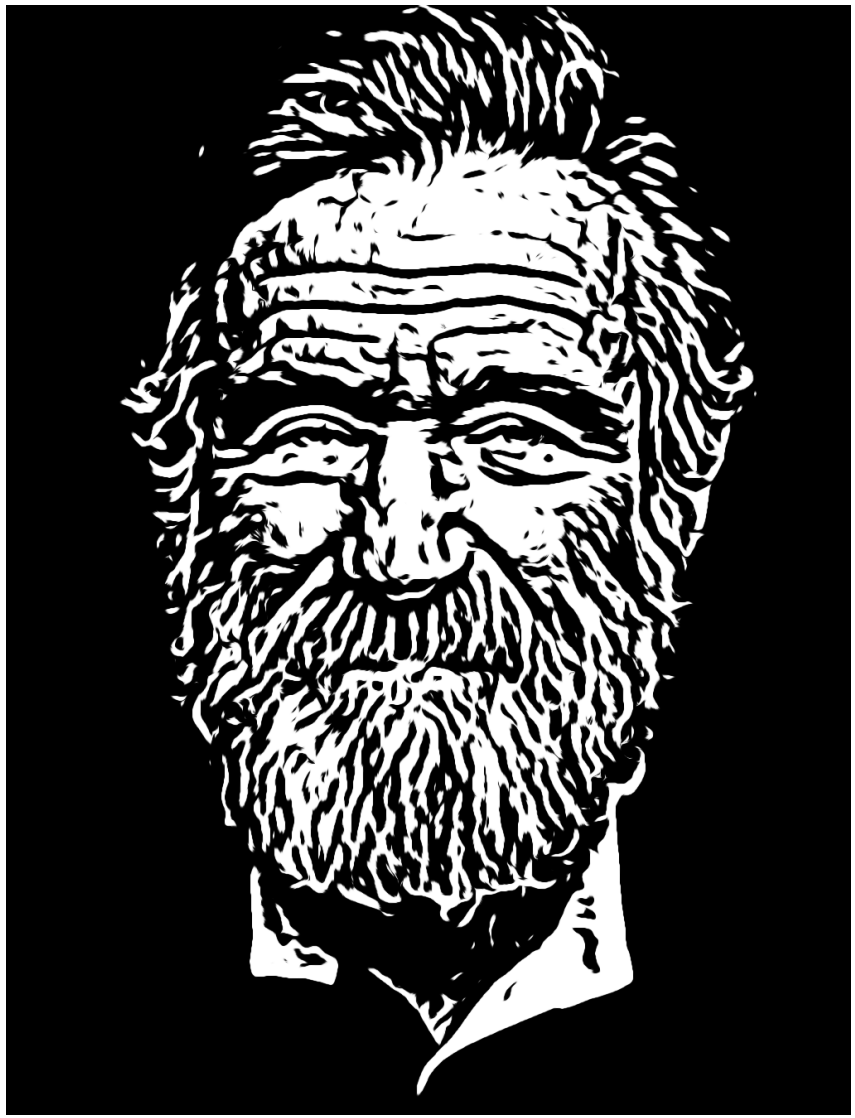
Exemplos - Carvão



Diferença das Gaussianas Baseada em Fluxo

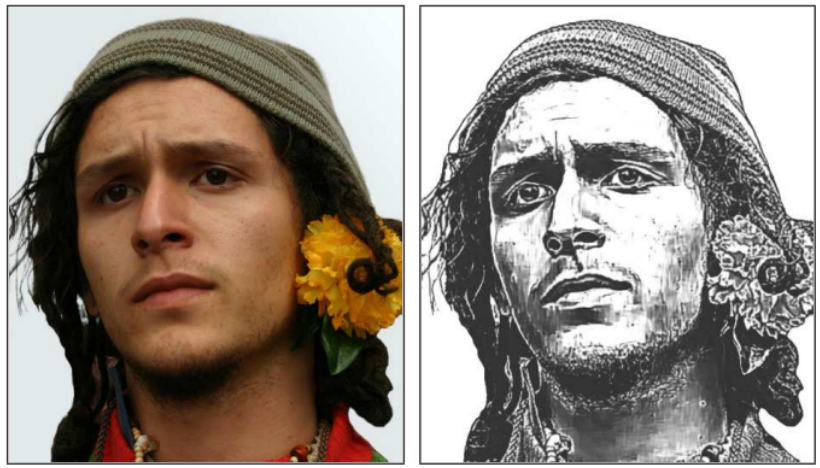
Exemplos - Xilogravura





Diferença das Gaussianas Baseada em Fluxo

Exemplos - *Cross-hatching*



(a) Source (b) XDoG

(c) XDoG Thresholding (d) XDoG-Hatching

Referências

Artigos

[1] Holger Winnemöller, Jan Eric Kyprianidis e Sven C. Olsen. «XDoG: An eXtended difference-of-Gaussians compendium including advanced image stylization». Em: *Computers & Graphics* 36.6 (2012). 2011 Joint Symposium on Computational Aesthetics (CAe), Non-Photorealistic Animation and Rendering (NPAR), and Sketch-Based Interfaces and Modeling (SBIM), pp. 740–753. DOI: 10.1016/j.cag.2012.03.004.

Links

[2] Acerola. *This is the Difference of Gaussians*. YouTube. URL: https://www.youtube.com/watch?v=5EuYKEvugLU.

[3] Tiago Maia de Lacerda Brasil. *analise-imagens*. GitHub. URL: https://github.com/TiagoLacerda/analise-imagens.

[4] Tiago Maia de Lacerda Brasil. *Difference of Gaussians*. URL: https://dog.tiagomlbrasil.com/.